Anup Sedhain

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Experience

Software Engineer, Airvet Inc. – Los Angeles, CA — May 2025 – Present

- Led development of Airvet's parent-facing web application using NextJs, Golang, Twilio, and AWS and spearheaded research into veterinary AI diagnostic tools leveraging RAG architecture with Langchain.
- Architected AI-driven automation using Model Context Protocol (MCP) to streamline end-to-end regression testing workflows, reducing QA time by 80% and accelerating feature-to-release cycles from 4 weeks to 2 weeks.
- Engineered end-to-end automated development pipeline integrating JIRA, Figma, and GitHub with MCP agents, enabling automated releases while maintaining 99.9% deployment success rate.

Graduate Research Assistant, University of Florida – Gainesville, FL — Jan 2025 – Jun 2025

- Developed a multistage social analysis pipeline processing 2M+ BlueSky events on the HiPerGator HPC cluster, combining BERT, K-means and TF-IDF for 92% accurate AI content detection with temporal network analysis (NetworkX, PageRank) to model attitude evolution.

Software Engineer, Airvet Inc. – Los Angeles, CA — May 2021 – Jul 2024

- Engineered a real-time Care Portal with React, Redux, and Golang, and optimized system performance using Server-Sent Events and Redis pub-sub; enabled vet-techs to handle 100+ concurrent cases, accelerating response times by 75% while maintaining sub-200ms latency and 99.9% uptime.
- Spearheaded the modernization of airvet.com, collaborating with product, marketing, and design to replace a legacy system with a Next.js application, leveraging WordPress as a headless CMS and implementing ISG.
- Owned the feature roadmap for key B2B products by building an analytics ecosystem (Firebase, Amplitude, LogRocket) to inform development; led the creation of new admin tools that empowered teams with data insights, directly contributing to product iterations that resulted in almost 100% client retention.

Software Engineer, Freelance, Polydelic AS – Oslo, Norway — Aug 2020 – Jun 2021

- Delivered 6+ SaaS platforms using React, Node.js, GraphQL, and AWS through cross-functional collaboration with product and design team, helping secure millions in funding and serving 500K+ users with 99.9% uptime.
- Engineered a video streaming platform by implementing a mediasoup-based Selective Forwarding Unit (SFU), achieving scalability to support 100+ concurrent users through strategic load balancing and Redis caching.
- Launched artist booking marketplace (Rubato) utilizing ElasticSearch and automated contract processing with GraphQL APIs, facilitating €100K+ monthly transactions.

Software Engineer, Leapfrog Technology Inc. – Kathmandu, Nepal — Jul 2019 – Dec 2020

- Spearheaded enterprise project management platform development using Firebase and LitElement with automated workflows, accelerating delivery timelines by 40% for 50+ projects.
- Helped build bio-informatics application managing protein workflows with MySQL database and Micronaut, supporting 200+ batches monthly and improving processing efficiency by 25%.

Education

University of Florida, Master of Science in Computer Science, GPA 4.0 — Aug 2024 – Dec 2025

Kathmandu University, Bachelor of Computer Engineering, GPA 3.63 — Sep 2015 – Jul 2019

Technical Skills

AI/ML: MCPs, RAG, Langchain, Vector Databases, Hugging Face, Ollama, TGNNs, ML Pipelines
Programming: Python, TypeScript, JavaScript, Go, C++, Node.js, Java, Rust, OpenMP, MPI
Frontend: React, Next.js, Redux, Tailwind CSS, WebRTC, WASM, Real-time Systems
Backend & Data: PostgreSQL, MongoDB, Redis, Firebase, ElasticSearch, Spring Boot, RESTful APIs, GraphQL
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Kafka